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Administrative Report Quarterly & Annual System Documentation

Controller Overview

ASP.NET Core Controller for managing quarterly and annual administrative reports with activity tracking and signature verification.

Property	Value
Area	Admin
Authorization	AdminAuthorize
Dependencies	IAdministrativeReportQuarterlyAnnualService, IMapper (AutoMapper)

Action Methods

Method	Purpose	Parameters	Features
Index	Main dashboard for quarterly/annual reports	selectedYear (optional)	Year dropdown population, Quarterly data aggregation, Activity and administrative counts calculation, Collaborative report signature checking, AJAX partial view support
All2Details	Combined administrative and activities details view	selectedYear, quarter, pagination	Delegates to service layer, Signature verification, Returns mapped data
AdminstrativeDetails	Administrative-only reports view	selectedYear, quarter, pagination	Quarter filtering, Annual summary (quarter=5), Signature verification
ActivitiesDetails	Activities-focused reports with chart data	selectedYear, quarter, pagination	Pie chart data, Monthly participation, Activity duration, Quarter-based grouping

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Method	Purpose
PrintActivitiesDetails	Printable version of activities report
PrintAdminstrativeDetails	Printable administrative reports
PrintAll2Details	Printable combined reports

All use [IgnoreAction] attribute to exclude routes.

Signature & OTP System

Feature	Description
SendOtp	Generates and sends OTP for specific roles (trainer/manager)
ValidateOtp	Verifies OTP codes for report signing with [NoLogging] attribute
Signature Integration	Signature checking across all detail views, Role-based authorization, Quarterly report type differentiation

Quarter Handling Code

```
List<List<int>> quarters = new() {
    new List<int>{1,2,3},    // Q1
    new List<int>{4,5,6},    // Q2
    new List<int>{7,8,9},    // Q3
    new List<int>{10,11,12} // Q4
};
```

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DTO Name	Level	Contains Details?	Primary Use
MonthlyAdministrativeReportDetailDTO	Item level	Yes	Represents one recorded activity
MonthsOfYearsAnnualyDTO	Monthly summary	No	Dashboard counts and summaries
MonthsOfYearsAnnualyWithDetailsDTO	Monthly detailed	Yes	Full reporting and printing
MonthsOfYearsAnnualyActivitiesAndAdministrativeDTO	Quarterly/Annual comparison	Yes	Charts, side-by-side analysis

ViewModels Overview

ViewModel	Purpose
ActivitiesReportQuarterlyVM	Quarterly activities report with detailed months and chart data
AdministrativeReportQuarterlyAnnualIndexVM	Index page summary counts and quarters dropdown
AdministrativeReportQuarterlyAnnualVM	Detailed ViewModel for quarterly/annual report
MonthsOfYearsAnnualyActivitiesAndAdministrative	Combines admin and activity data for quarterly charts
MonthsOfYearsAnnualyVM	Represents monthly data within a year
MonthsOfYearsAnnualyWithDetailsVM	Monthly data with child details and paging
OTPRequest	OTP validation request for signing reports
PieChartVM	Stores chart data for quarterly participation visualization
QuartersReportActivitiesDetailsVM	Detailed activities for a specific quarter report
QuartersReportAdminstrativeDetailsVM	Detailed administrative report for a specific quarter
QuartersReportCollaborativeDetailsVM	Collaborative reports combining activities and administrative data

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Feature	Description
Quarter Management	Dynamic availability, filtering, validation, empty state handling
Signature Workflow	Manager signature requirement, OTP-based authentication, role-based access
Data Calculation	Activity duration summation, participant aggregation, monthly subtotals, grand totals

Technical & Integration Points

Feature	Description
Charts	Chart.js, Doughnut/Bar charts, Multi-level, Color coding, Custom center text, Tooltips
State Management	Session storage, AJAX partial updates, Modal management
Export	Print functionality, Signature requirement enforcement, Year/Quarter parameter passing
Localization	Resource files for multi-language support, Culture-specific loading